

Maximizing Power Generation and Stiffness to Mass Ratio of FDM Printed Tabletop Wind Turbine

Group 15: ¹Shuqing Zhao, ¹Preston Bhat, ¹Arjun Narayanan, ¹Krishna Ram, ¹Deynna Quintero, ¹Kina Li

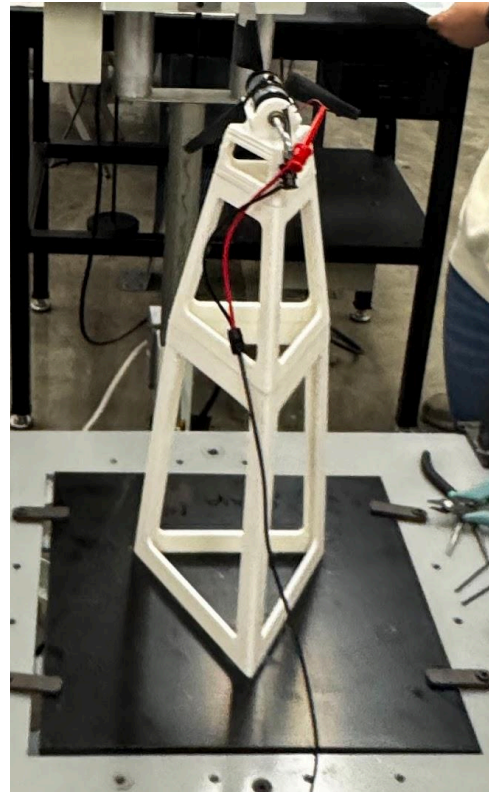
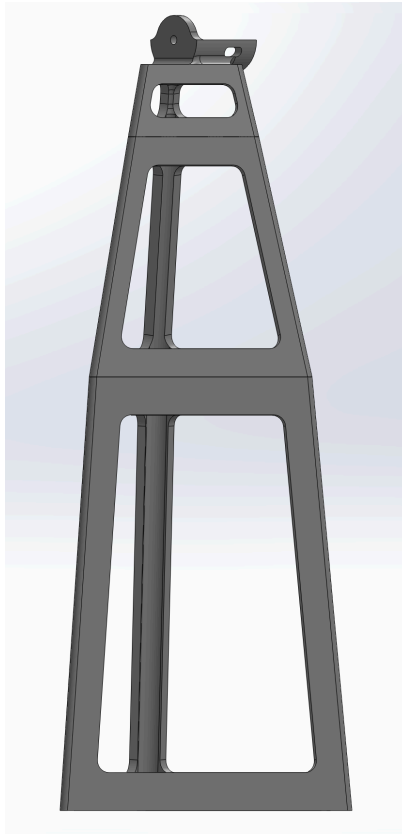
Professor Kourosh Youssefi

¹All authors contributed equally

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Project Summary



Figure 1. Turbine during testing for power generation.

This project consisted of designing, prototyping, and testing a small-scale wind turbine. We were able to showcase our knowledge of CAD through the program SolidWorks, 3D printing, and the ability to carry out physical testing to assess our turbine's efficiency. The aim was to produce and manufacture a turbine capable of generating, ideally, over 2 watts of power, staying under a weight of 300g and having the highest stiffness possible. The volume of this tower couldn't exceed 17 cubic inches. Through a combined effort, we managed to create a small-scale turbine that satisfies many of these goals and all of the requirements.

The first piece of the turbine we designed was the blade, which we focused on having the most airflow and the lowest amount of drag. This led the team to create a blade inspired by the SG6043 airfoil since it works best on more minor scales. Since we expected winds of about 25 mph, we used three blades with an angle of twist that started at 10 degrees and twisted to 5 degrees at the tip. This fulfilled the blade requirements and ensured satisfactory results. Then we designed the tower, which had to stay under a height of 16 in. At first, the design was a triangular shape with helical supports going through the tower's bottom piece. When creating the 3D model of this design, it exceeded the volume requirements, which led the team to create a simplified version without the helical supports and split our tower into three pieces to fit the 3D printer's height requirements.

The results of our tower during testing were adequate. Our tower has a mass of about 305.3 grams after gluing but initially weighed 295.2 grams as individual components after printing. The stiffness resulted in 17.755 N/mm with a linear correlation coefficient value of 0.987, and the maximum power generated was 1105.0 mW. We had an efficiency of 14.1%. The results were lower than anticipated but were still within an acceptable range. Figure 1 shows the tower during testing with the blade and motor attached.

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Introduction

In the face of climate change, resource depletion, and growing global energy demands, pursuing sustainable energy alternatives has become one of the most critical challenges of our time. Fossil fuels—long the dominant energy source for industrialized societies—are finite and produce significant amounts of greenhouse gas emissions. These emissions contribute to global warming, air pollution, and environmental degradation. As a result, there has been a major global push toward decarbonizing energy systems by adopting clean, renewable sources such as solar, hydro, geothermal, and wind energy. Among these, wind energy stands out due to its scalability, technological maturity, and relatively low environmental footprint during operation (Manwell, McGowan, & Rogers, 2010).

Wind energy is generated by converting the kinetic energy of moving air into mechanical rotation, which is then used to drive an electrical generator. The fundamental physical principle is based on the aerodynamic interaction between the wind and turbine blades, which induces rotation in the rotor. This mechanical energy is transferred to a generator via a shaft, where it is transformed into usable electrical energy. This process is highly efficient under optimal conditions and can be deployed in both centralized (utility-scale wind farms) and decentralized (small-scale or off-grid) systems. As of 2023, global installed wind power capacity has surpassed 900 gigawatts, a figure that continues to grow each year due to supportive policies, technological innovations, and cost reductions in turbine manufacturing and maintenance (Global Wind Energy Council, 2023).

The idea of capturing wind energy is far from new. Evidence of wind-powered machines dates back over 1,000 years. Ancient Persian and Chinese societies used early windmills for

irrigation and grain processing. In medieval Europe, windmills became a cornerstone of agricultural economies, particularly in the Netherlands and Britain, where they were used to pump water and grind flour. Although mechanically simple, these early designs demonstrated the practical benefits of wind as a reliable energy source (Manwell et al., 2010). Over time, these devices evolved from horizontal-axis wooden structures to modern wind turbines made of composite materials, precise aerodynamic profiles, and sophisticated control systems (Hau, 2013).

Modern wind turbines can be broadly categorized into horizontal-axis wind turbines (HAWTs) and vertical-axis wind turbines (VAWTs). HAWTs are the most common configuration and feature blades that rotate around a horizontal axis, typically oriented into the wind using yaw mechanisms. These turbines are often mounted on tall towers to access stronger and more consistent winds at higher elevations. VAWTs, while less common, offer advantages such as simpler design and the ability to operate at lower wind speeds or in turbulent environments (Ackermann, 2005). Regardless of type, modern turbines integrate multiple subsystems—blades, nacelle, gearbox, generator, controller, and tower—each of which must be carefully engineered to ensure safety, durability, and efficiency under varying environmental conditions.

Wind turbines are now deployed in a wide range of environments. Large-scale wind farms are commonly found in flat, open areas such as the American Midwest, coastal regions, and increasingly in offshore settings where wind speeds are higher and less intermittent. Offshore wind turbines, in particular, represent a fast-growing sector due to their higher capacity factors and reduced land-use conflict (GWEC, 2023). In contrast, small-scale wind turbines are typically used in rural or isolated communities, where they serve as stand-alone power systems or are combined with solar panels and battery storage in hybrid microgrids. These diverse

applications demonstrate the adaptability and importance of wind energy across a range of scales and environments. While utility-scale wind farms dominate global capacity, small-scale wind turbines play a critical role in research, education, and remote energy access. They offer a practical platform for exploring the core engineering principles behind wind energy systems, such as aerodynamics, mechanical design, and energy conversion efficiency. In this context, our project focuses on designing, fabricating, and testing a small wind turbine, giving us the opportunity to apply theoretical knowledge to a hands-on engineering challenge while gaining a deeper understanding of sustainable energy technologies.

In this project, we explored these principles by designing and fabricating our small-scale wind turbine, applying engineering concepts through hands-on design, prototyping, and testing. The main goals of this project were to apply computer-aided design (CAD), use additive manufacturing (3D printing), and perform physical testing to evaluate our turbine's mechanical and electrical performance. Additionally, the system's ability to convert wind energy into electrical power and to withstand structural deflection were experimentally tested and analyzed. By going through the entire design-build-test cycle, we gained hands-on experience in structural design, aerodynamics, materials limitations, and power generation.

Our wind turbine design consisted of three main components: the rotor blades, the support tower, and the generator housing (included in the support tower's upper platform). We were required to create all parts using CAD software (SolidWorks) and fabricate them entirely using PLA plastic through 3D printing. The project included several design constraints to simulate real-world engineering challenges. For example, the turbine rotor blades were limited to a maximum radius of 3 inches from the hub center, ensuring a swept area diameter of no more than 6 inches. The support tower had to be fabricated in multiple pieces to comply with the 3D

printer's nine-by-nine-by-nine-inch volume constraint, and its material volume could not exceed 17 cubic inches. The entire assembly had to maintain radial symmetry and structural stability with at least three contact points on the provided lower platform. These constraints encouraged thoughtful engineering decisions, especially in optimizing blade aerodynamics, tower stiffness, and ease of assembly.

To design our wind turbine rotor, we considered both aerodynamic efficiency and structural simplicity. We aimed to develop a blade shape that would generate high rotational torque under a wind speed of approximately 25 mph while ensuring smooth and stable operation. Figure 1 below illustrates two key aspects of our blade design: the aerodynamic top profile and the symmetric 3-blade top view. The design was further informed by concepts such as angle of attack, tip-speed ratio, and Reynolds number—all of which influence how effectively the blades interact with the airflow. Our blade design was centered around achieving optimal aerodynamic performance at low Reynolds numbers, which is typical for small-scale wind turbines like ours.

We selected a blade profile inspired by the SG6043 airfoil, which is specifically optimized for low Reynolds number conditions (approximately 30,000–50,000), excelling in laminar flow. This airfoil offers an excellent lift-to-drag ratio, which is critical for maximizing torque and rotation efficiency under modest wind speeds. The SG6043 profile features a maximum thickness of 10% located at 32.1% of the chord and a maximum camber of 5.1% at 53.3% of the chord. These proportions give the blade a moderately curved upper surface and a thinner trailing edge, promoting smoother airflow and reducing drag. Our blade shape incorporated these features in a simplified form to ensure manufacturability while still capturing the core aerodynamic benefits. To determine the most effective blade operation, we considered the angle of attack, which is the angle between the chord line of the airfoil and the relative wind

direction. For our expected Reynolds number range ($\sim 30,000$), previous studies and polar plots suggest that the optimal angle of attack falls between 8° and 10° , which gives the best lift-to-drag ratio without inducing early flow separation or stall (Airfoil Tools, 2024). However, since the angle of attack varies along the radius of the blade due to changes in relative wind speed, we implemented blade twist into our design. The twist was applied along the length of each blade, transitioning from approximately 10° at the root to 5° at the tip. This adjustment ensures that each blade section operates near its optimal angle of attack despite varying local flow conditions. The twist helps maintain high aerodynamic efficiency across the blade span and avoids conditions leading to a stall at either end. This approach is commonly used in real wind turbine blades to equalize performance across the entire airfoil surface (Youseffi, 2024). We also confirmed our decision to use three blades for the turbine. This configuration offers a strong balance between power output and structural simplicity. Using three blades improves aerodynamic capture while keeping the rotational system balanced and mechanically stable. As noted in class discussions and technical literature, an even number of blades can lead to dynamic imbalances and reduce efficiency (Hau, 2013).

By combining these principles with practical considerations like manufacturability and print size constraints, we arrived at a blade geometry that was theoretically sound and feasible for 3D printing.

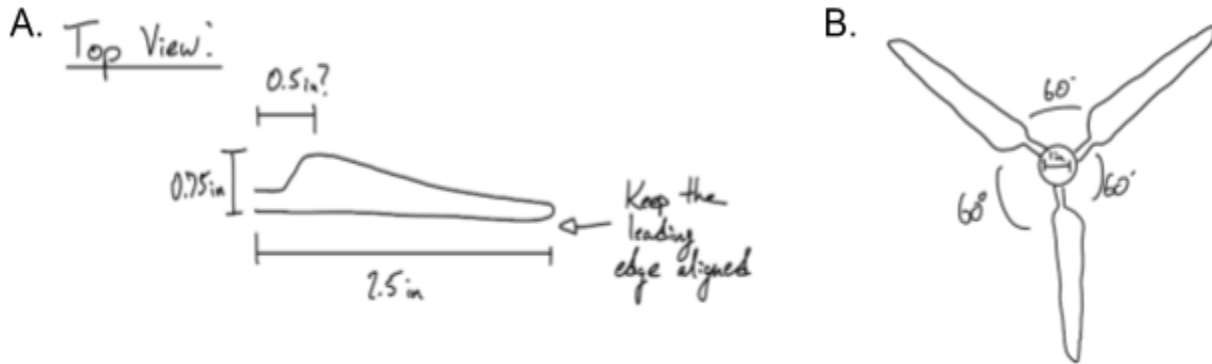


Figure 1. Combined sketch of blade design: (A) top profile inspired by a low-speed airfoil; (B) 3-blade top view with equal 120° spacing around the hub.

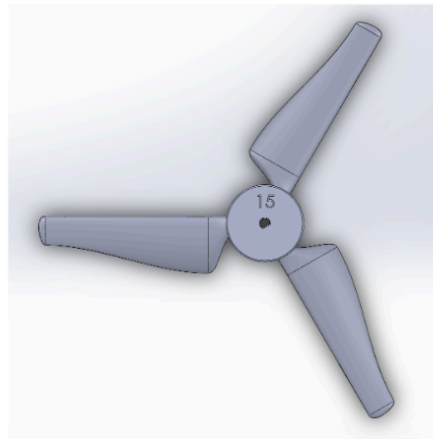


Figure 2. CAD of turbine blade with desired features which was ultimately printed and tested.

At the same time, our wind turbine tower was designed to focus on both structural stability and printability while also ensuring it met the specific dimensional and volume constraints outlined in the project requirements. Figure 3 below illustrates the conceptual sketch of our tower design, including annotations on geometry, loading, and fabrication strategy.

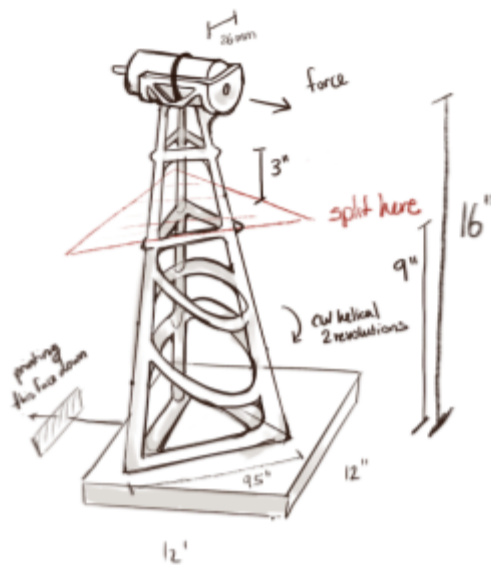


Figure 3. Freehand sketch of the preliminary tower design, annotated with dimensions, support geometry, and key structural considerations.

Prior to testing, we revised our original tower design to comply with the project's material volume requirement of 17 cubic inches, as our initial structure exceeded this limit. Figure 4 shows the modified tower that was ultimately fabricated and tested.



Figure 4. CAD of final tower design updated to fit into dimensions and weight requirements.

The original design featured more complex geometry, including helical or curved supports, preserving structural integrity while ensuring the design could be printed efficiently. We incorporated a clockwise (CW) helical support structure wrapped around the tower's lower portion to improve torsional rigidity and reduce flexing, especially under lateral loads during operation. This design was inspired by truss-based architecture, where distributed load paths help prevent buckling and resist deformation under force. Additionally, the open-frame design reduces material use, which we thought would help us stay within the 17 in³ volume limit while offering strong vertical and lateral support. The flat surface underneath the cradle ensures a consistent force application during loading. To ensure proper adhesion during 3D printing and reduce the likelihood of support failure or warping, the design was oriented face-down on the print bed, as noted in the sketch.

Upon running a SolidWorks mass properties analysis, we discovered that in our attempt at increasing stiffness, the total print volume was too high and would disqualify the design from meeting the required constraints. As a result, we simplified the structure to reduce print volume while preserving overall mechanical stability and resistance to deformation when enduring wind forces and load testing. The updated design replaced intricate internal geometry with more minimalist, angular trusses that use less material but still provide sufficient support. This permitted us to stay within the allowed volume while still meeting the necessary height (16 inches) and footprint (12" × 12") specifications. We also preserved the split feature for 3D printing compatibility, dividing the tower into two parts to fit within the 9" × 9" × 9" build envelope of the printer. Although this modified version was not structurally tested prior to fabrication, we made design decisions based on theoretical considerations such as load path

alignment, triangular bracing, and overall symmetry to help ensure fundamental mechanical integrity.

These choices were intended to balance printability, compliance with project rules, and functional performance during later stiffness and power generation tests.

From a functional standpoint, the wind turbine was evaluated based on two key criteria: its ability to generate electrical power and its mechanical stiffness. Power output was measured using a load box and digital power meter setup, and maximum power was determined by varying electrical resistance to identify the optimal loading condition. This process reflects real-world practices where turbines are coupled with smart controllers to optimize output based on changing wind conditions (Ackermann, 2005). The tower's stiffness was measured using a deflection test, where weights were applied to the top platform, and displacement was recorded using a dial indicator. The slope of the resulting load-displacement curve provided the stiffness value. These experimental results were then compared to theoretical predictions to assess overall design efficiency.

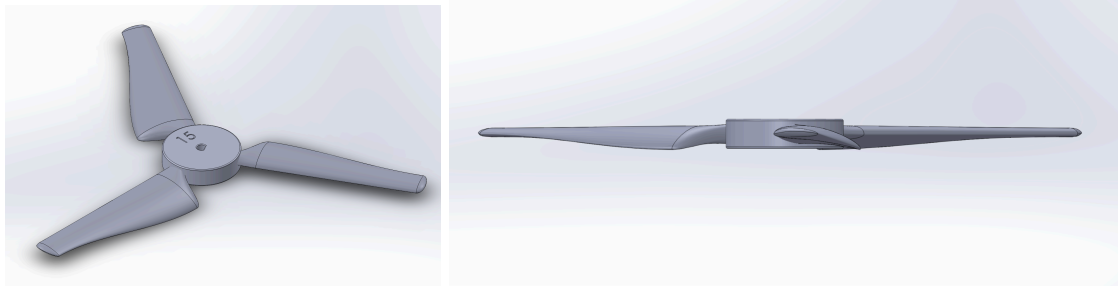
The specific objectives of our design were threefold. First, we aimed to maximize power generation efficiency through optimal blade design and loading conditions. This was achieved by rigorously evaluating blade performance factors and selecting a blade profile inspired by the SG6043 airfoil, which is well-suited for low Reynolds number conditions. Second, we sought to ensure the structural stability of the tower while remaining within strict volume and size constraints. To do so, we iteratively redimensioned and redesigned the tower, incorporating secure structural elements such as trusses and triangular frameworks. Third, we focused on developing a modular design that could be easily reproduced and tested using only the provided

specifications and equipment. Throughout the design process, we ensured that both the blade and tower could be 3D printed without requiring excessive support material, enabling efficient and consistent fabrication.

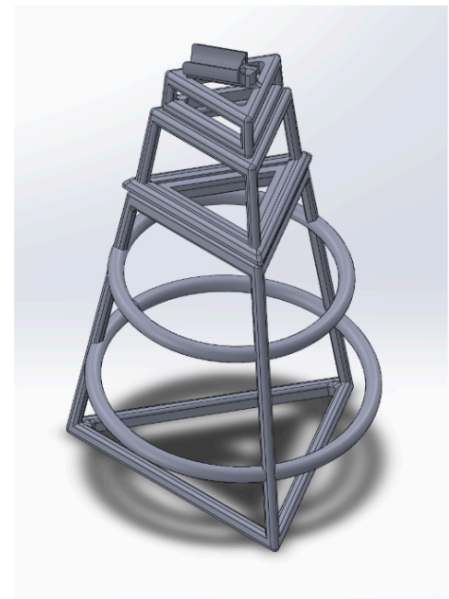
This report details our design approach, CAD modeling process, fabrication techniques, testing procedures, and results. The report includes sketches, diagrams, photographs, and data plots to help communicate our methodology and outcomes. This project exemplifies the balance of creativity, constraint-based design, and performance optimization that characterizes real-world engineering challenges.

Design

The rotor design features three blades attached to the hub, spaced even 120 degrees apart. The blades achieve a length of two and a half inches, with the hub diameter being one inch across, resulting in a six-inch circle while spinning. The maximum width of the blade is 0.75 inches before shorting as it reaches the hub. Many of the design decisions are articulated in the prior section, but for the sake of consistency, the angle of attack is 10 degrees, and the twist is 10 degrees at the root and slowly reduces until it reaches 5 degrees at the tip, as seen below. Embossed on the hub is our group number.

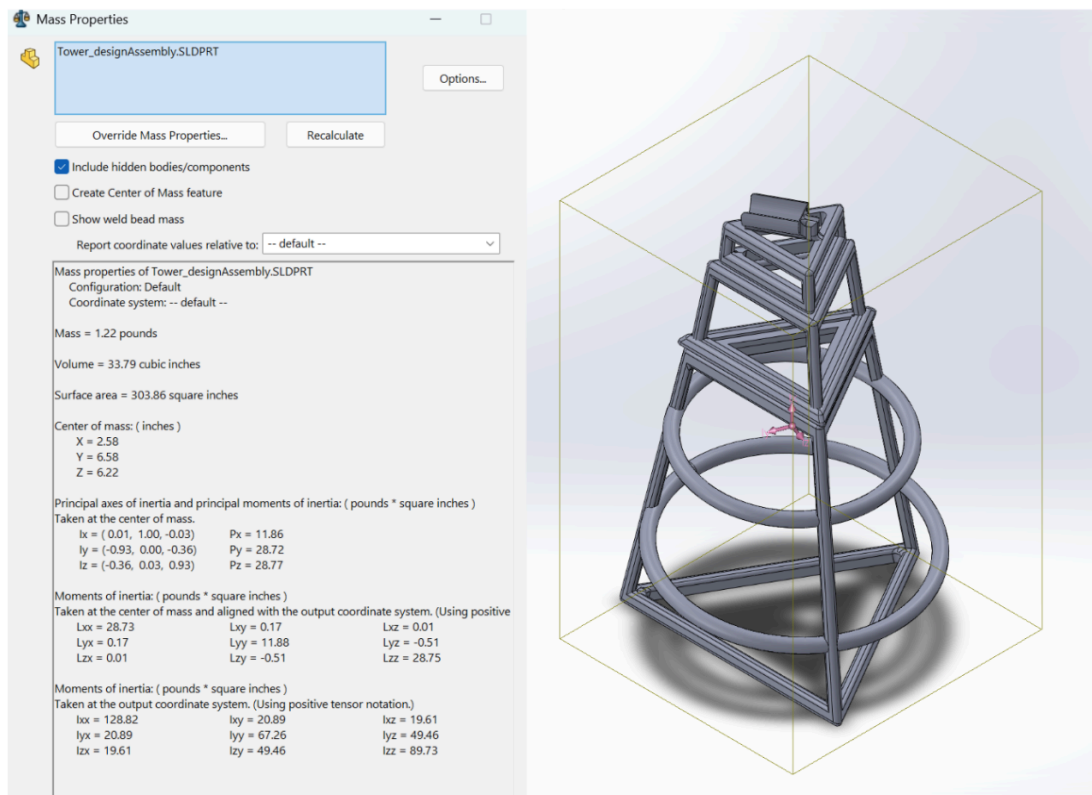


We designed the initial tower structure to be 16 inches tall, as required to align the generator shaft with the correct elevation from the base. To stay within the 3D printer's 9×9×9 inch build volume, we split the tower into three main sections: a lower base segment (approximately nine inches tall), a middle section (five inches), and an upper platform segment (two inches). The split occurs just below the platform that holds the motor, and the two parts are joined during post-processing using adhesive. The



helical rib makes two complete revolutions around the tower body and is anchored at multiple

points. The motor sits in a rectangular cradle at the top of the tower, sized to accommodate the 26 mm diameter motor housing, with an embedded 3/16-inch hole at the rear for inserting an eyebolt during the stiffness test. The entire assembly is mounted on a 12×12 inch ABS base platform provided in the project kit. The tower's footprint tapers inward toward the top, helping to maintain a low center of mass and increasing stability during wind testing. This orientation provided a flat contact surface and minimized the need for excessive support material. However, this design proved to be too ambitious, as the volume was almost double the 17 cubic inch limit. We modified the design to get our final turbine structure under the limit.



Our final turbine structure is based on a triangular frame on three different planes, supported by and connected via a set of angular trusses. To meet volume constraints, minimize mass, and reduce the impact of wind forces, it has significant extrusion cuts, relying on thin but sturdy trusses to carry the load from the tower. As mentioned in the prior section, this version of

the tower was also printed in three separate pieces of nine, five, and two-inch heights, ensuring that we did not exceed the maximum printable height (nine inches) in the complete 16-inch structure. In order to make sure that the tower did not suffer from failures related to assembly, we utilized the triangles to maximize the gluing contact area between each component and the base, as well as evenly distribute the load from the tower across each of the three sides. Instead of mounting the motor housing separately, we built it into the top section of the tower, reducing mass but also increasing the contact area with more structurally sound parts of the tower. With these practices implemented, our tower met the volume constraint but was on the heavier side, coming in with a mass of 493.2 grams. To accommodate this, we utilized 60% infill during our 3D print, getting us down to 295.2 grams prior to gluing/assembly.

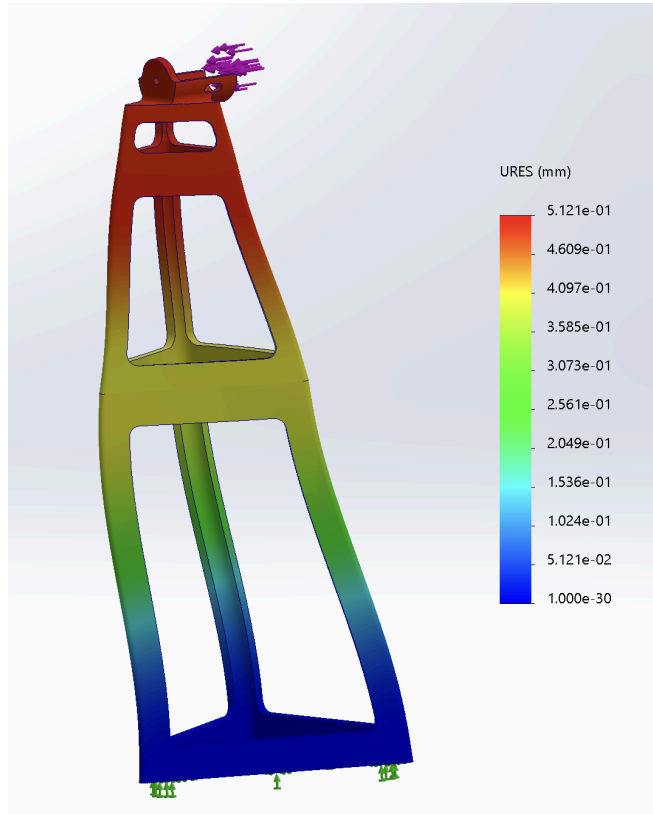
We utilized SolidWorks throughout our tower and blade design process. SolidWorks is a Computer-Aided Design, or CAD, software that allows users to design three-dimensional models, create two-dimensional drawings for the models, and test the models using structural studies known as Finite Element Analysis. All three of these components of SolidWorks were used throughout the course of this project, ensuring a well-thought-out, ambitious design that could both be tested to ensure structural soundness and effectively communicated for final production.

In the following simulations, we had to give the PLA of the tower custom properties, including yield strength, which we utilized as 35MPa based on the details from the manufacturer. In the FEA study, we bonded the three parts that had already been mated in the assembly to resemble the effect of the Loctite 435 adhesive and applied fixtures on the bottom face of the tower, as seen in the images below. We then applied a force of 1 kgf on the front face of the motor housing, mimicking the location of the force applied during the stiffness testing.

Custom PLA Values Based on Manufacturer Details

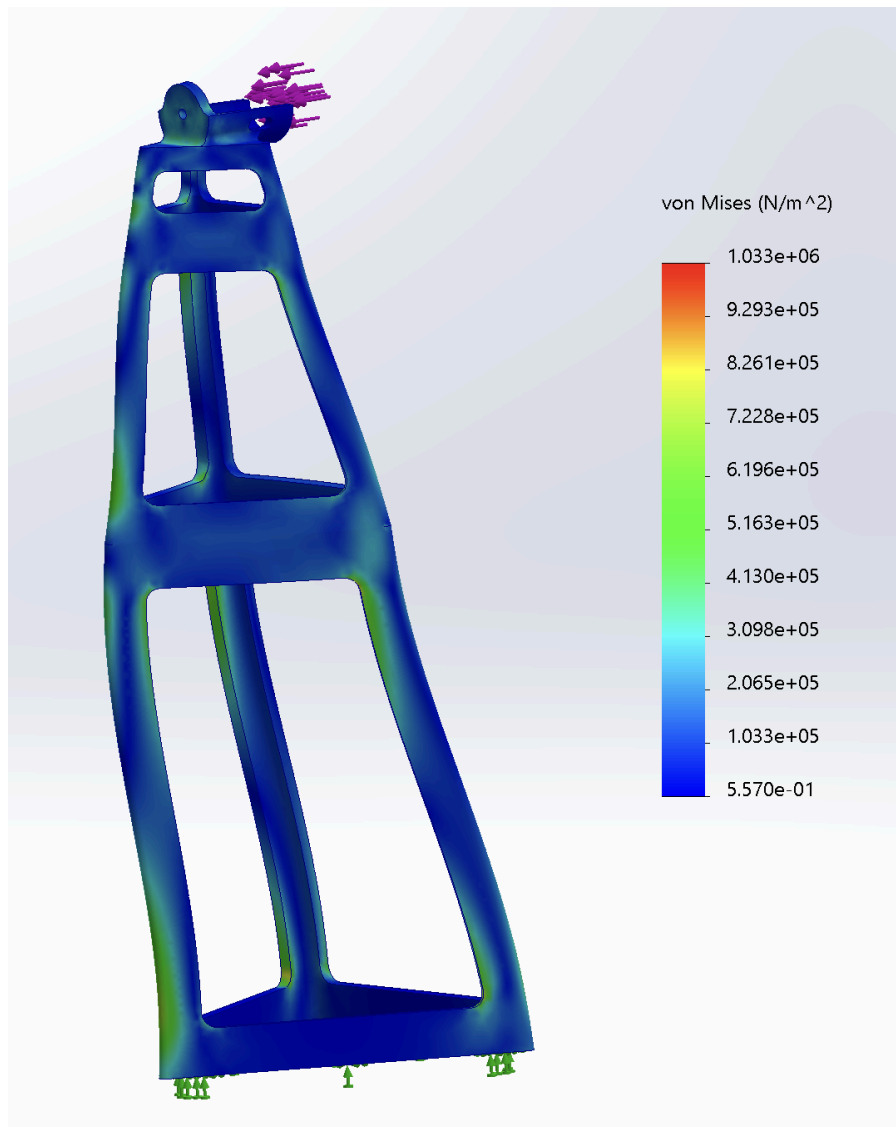
Elastic Modulus	Yield Strength	Poisson Ratio	Mass Density
1.27 GPa	35 MPa	0.394	1770 kg/m ³

Plot of Displacement



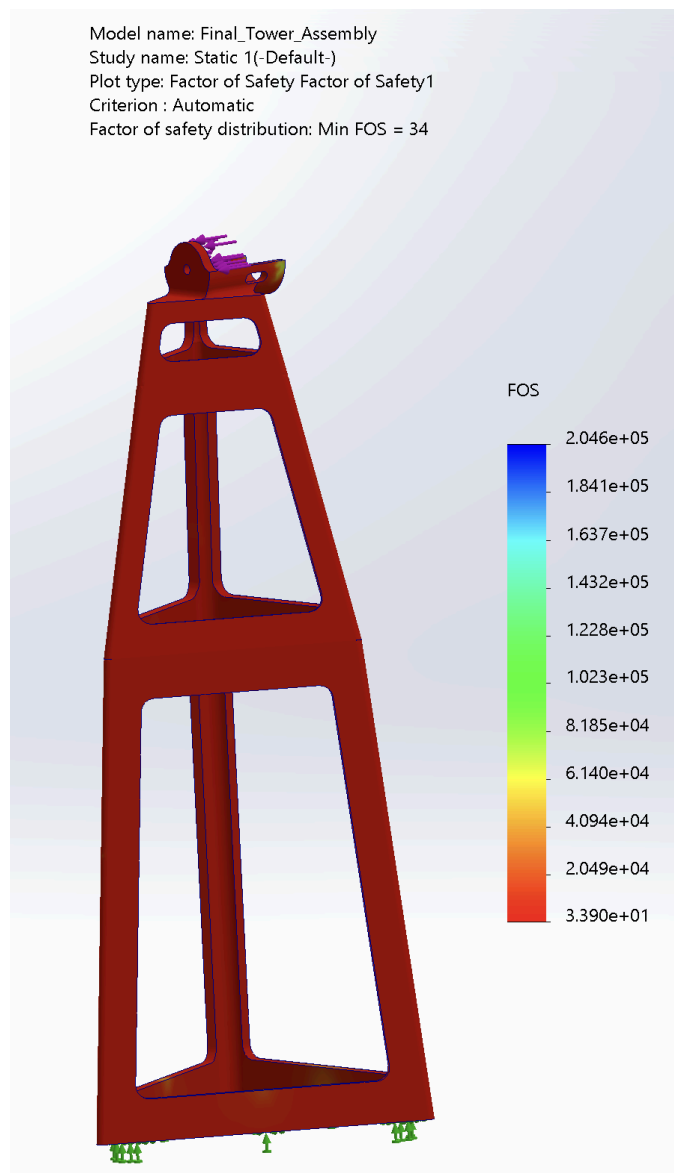
The plot represents the maximum displacement of the tower, given the loading aforementioned conditions. The simulation resulted in a displacement of 0.5121 millimeters, which was nearly identical to our experimental value of 0.52 millimeters. We believe we can attribute the slight discrepancy between these two values to either the difference between assembly bonds and Loctite 435 strength, a slight underestimate of the elastic modulus of the material, which we put at 1.27 GPa, or a combination of the two factors.

Plot of Stresses



This simulation shows the stress distribution under top loading. The maximum stress is about $1.03 \times 10^6 \text{ N/m}^2$, which is well below our PLA yield strength of $3.50 \times 10^7 \text{ N/m}^2$, confirming the design remains safely within the elastic range. The structure yields most on the edges of the struts, which makes sense given that those locations have the highest stress concentration, even with the fillets, in a load-bearing part of the structure.

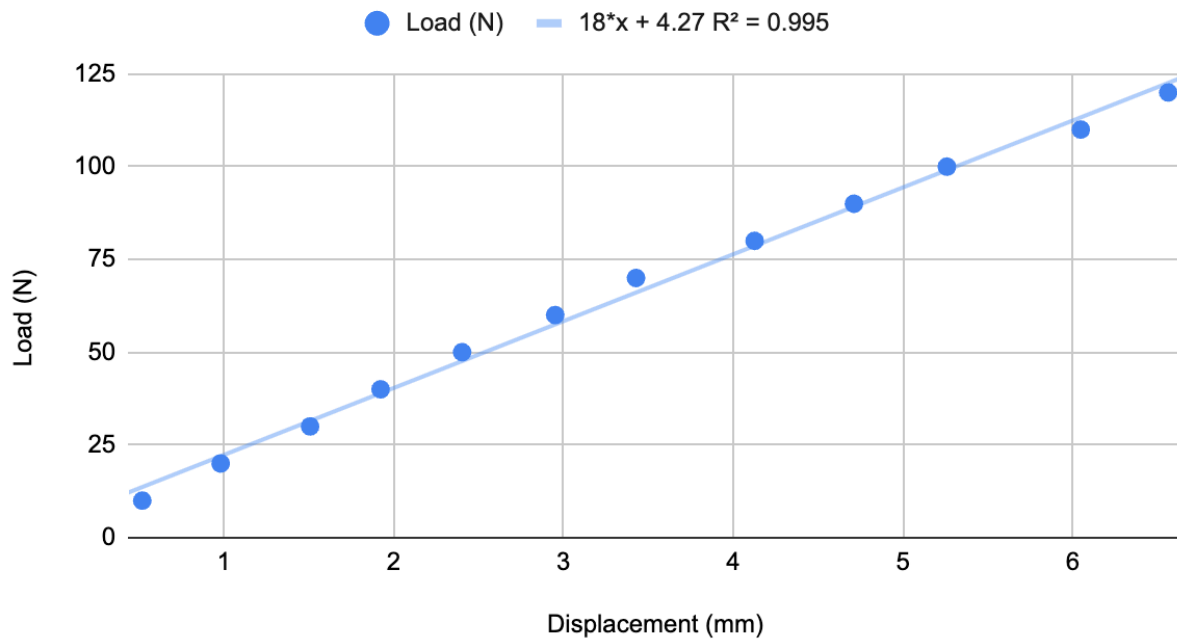
Plot of Factor Of Safety



This simulation shows the safety factor distribution of the tower under load. The minimum factor of safety is 34, meaning that the structure can handle 34 times the current load of 1 kgf before failing. This indicates that the tower is very structurally sound and operates well below material failure limits. We would not be able to get the tower to fail experimentally, as the maximum load available during tower testing was 50 kgf.

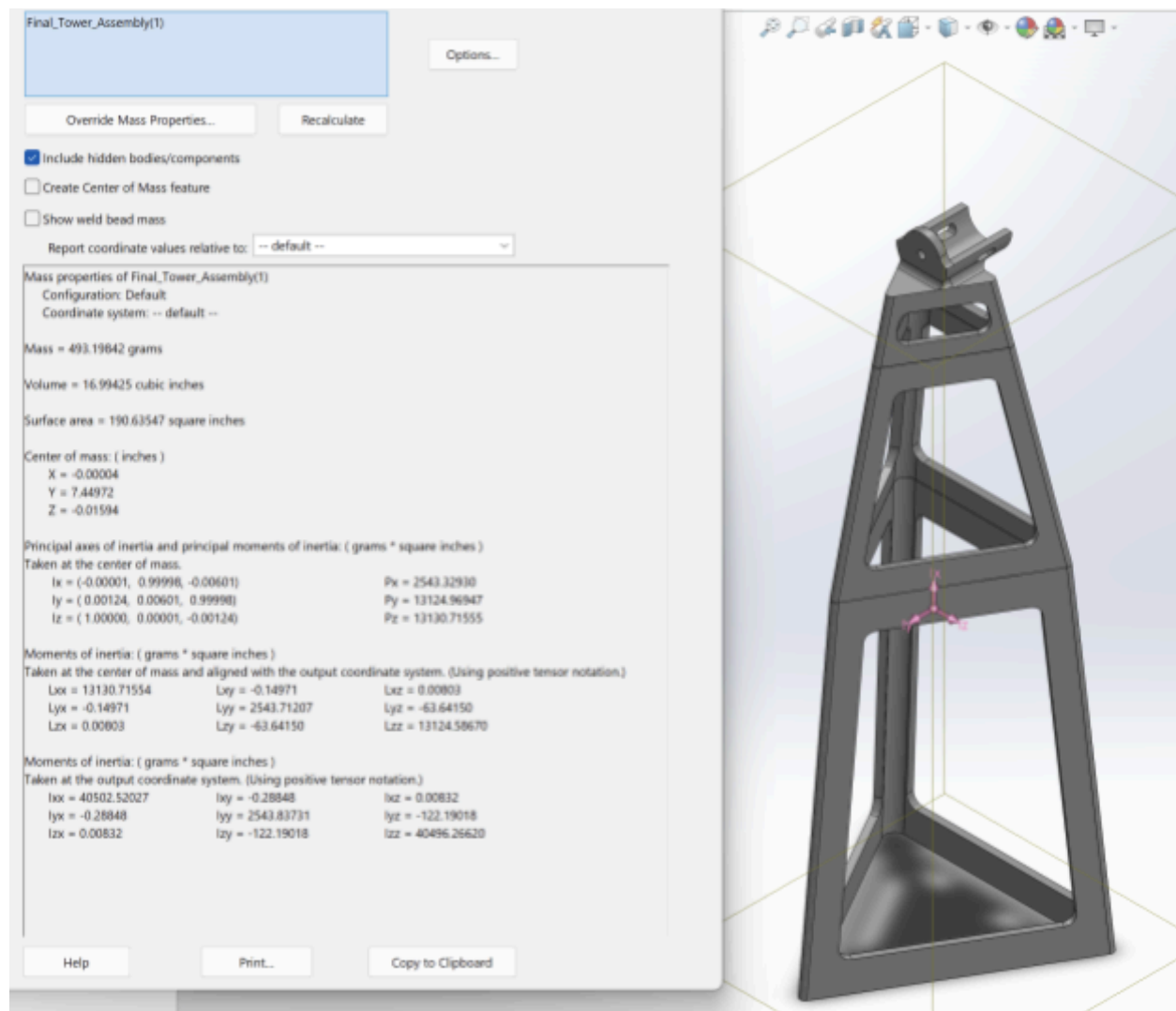
Load vs. Displacement Plot

Load (N) vs. Displacement (mm)



In this plot, we can see that there is a very strong linear relationship between the load and displacement, with a theoretical stiffness of 18 N/mm.

Mass Properties Window of Tower



Testing and Results

Equipment Used

Tachometer: Measures the angular velocity of our blade in rotations per minute



Power Meter (Load Box): Controls and gives a reading of the power in milliwatts, current in milliamperes, voltage in volts, and resistance in ohms of our wind turbine



Anemometer: Measures the wind velocity of the blower fan in miles per hour



Dial Indicator: Measures the deflection of the tower in millimeters



Testing Process for Power Generation Test

1. Setting Up the Turbine

- a. We started by securing the base plate of our wind turbine to the test platform.
Then, we attached the motor to the motor housing using zip ties to hold it firmly.
Finally, we pushed the turbine blades onto the motor by aligning the center hole.
2. Positioning the Blower
 - a. We placed a blower in front of the turbine and adjusted it so the wind speed reached about 25 mph. We used an anemometer to measure the exact wind speed.
3. Wiring the System
 - a. With the blower off, we connected the motor (acting as a generator) to the load box using IC hook grabber cables.
4. Running the Test
 - a. We turned the blower back on and slowly turned the potentiometer knob clockwise in small steps. At each step, we recorded the current (mA), voltage (V), power (mW), and resistance (Ohms) from the digital display on the load box in our data table.
5. Measuring RPM
 - a. At each step, we also used a tachometer to measure how fast the blades were spinning (RPM) by pointing it at a reflective tape on the blade. We added this data to our table as well.
6. Analyzing the Results
 - a. Using our recorded data, we plotted two graphs: Voltage vs. Current and Power vs. Current. We fit the data to identify the point where the turbine produced the most power.

Testing Process for Stiffness Test

1. Setting Up the Pulley System

- a. We screwed an eyebolt into the back of the motor housing and tied a string to it.
The string was looped over a pulley to allow us to hang weights from it.

2. Replacing the Motor

- a. To safely simulate loading, we placed a gray cylinder where the motor would go and secured it with the eyebolt.

3. Positioning the Dial Indicator

- a. We adjusted the dial indicator so it gently pressed into the cylinder and set it to zero.

4. Adding Weights and Measuring Deflection

- a. We gradually added weights in 100g increments. After reaching 1000g, we used larger increments of 1000g until we reached 4000g total. After each added weight, we recorded how much the tower deflected using the dial indicator.

5. Measuring Tower Mass

- a. We weighed our turbine model (without the motor or blades) and subtracted the base plate's weight to get just the tower's weight. We then locked the base plate onto the test platform.

6. Analyzing the Results

- a. We plotted a Load vs. Deflection graph based on our data, fitted a straight line to the points, and used the slope of that line to calculate the stiffness of the wind tower.

Table 1. Wind Turbine Tower Stiffness Test Data

This table presents the displacement measurements of the wind turbine tower under a series of vertical loads applied in 0.1 kilogram-force increments. These were then multiplied by 9.80 to convert the values to newtons. A dial indicator was used to record displacement in millimeters at the top of the tower. These values were used to plot load versus displacement, and the slope of the resulting curve was used to determine the tower's stiffness.

Data Points	Load (N)	Displacement (mm)
1	0.98	0.08
2	1.96	0.11
3	2.94	0.16
4	3.92	0.20
5	4.90	0.26
6	5.88	0.32
7	6.86	0.35
8	7.84	0.41
9	8.82	0.47
10	9.8	0.52
11	10.78	0.60
12	11.76	0.70

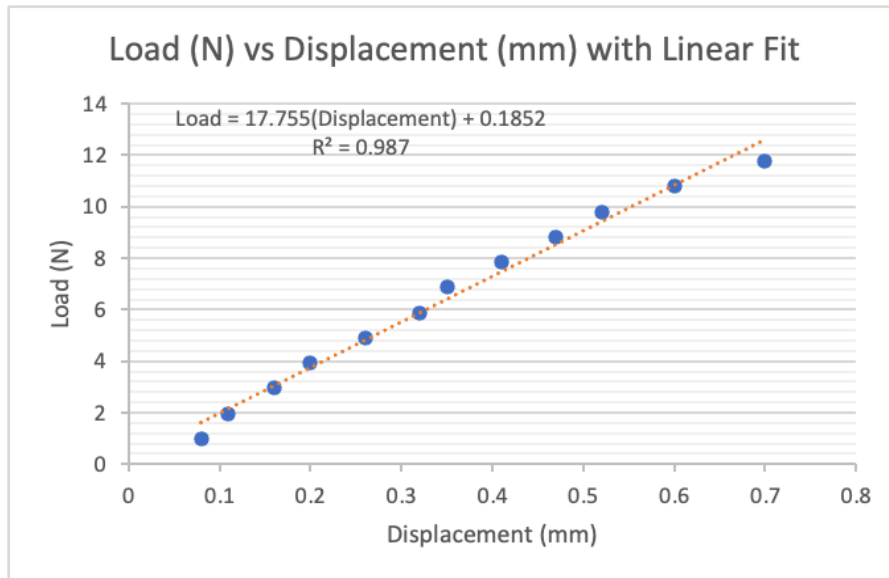
Table 2. Wind Turbine Power Generation Test Data

This table summarizes the electrical performance of the wind turbine at various load conditions. A power meter (load box) was used to vary the electrical resistance and record the corresponding voltage (V), current (mA), and power (mW) output. A tachometer was used to measure the blade's rotational speed (rpm), and an anemometer was used to verify wind speed during the test. This data was analyzed to identify the peak power output and calculate system efficiency.

Data Points	Resistance (Ohms)	Voltage (V)	Current (mA)	Power (mW)	Speed (rpm)
1	51	3.65	62.2	223.2	6194
2	45	4.13	85.0	358.9	5974
3	40	4.21	84.6	336.1	6125
4	35	4.02	103.4	427.5	5334
5	30	3.70	92.4	337.4	5335
6	25	3.64	122.8	436.1	5299
7	20	3.68	118.4	431.5	5367
8	15	3.54	165.9	596.6	5280
9	10	3.58	155.6	558.9	5331
10	5	3.28	263.1	951.4	5073
11	3	3.34	273.1	1105.0	5136
12	1	3.26	294.1	883.0	5045
13	0	2.95	360.1	878.3	4714

Plot 1: Load vs. Displacement

This plot shows how much the tower deflected (in mm) under different applied loads (in N). The linear region of this plot can be used to determine the stiffness of the tower.

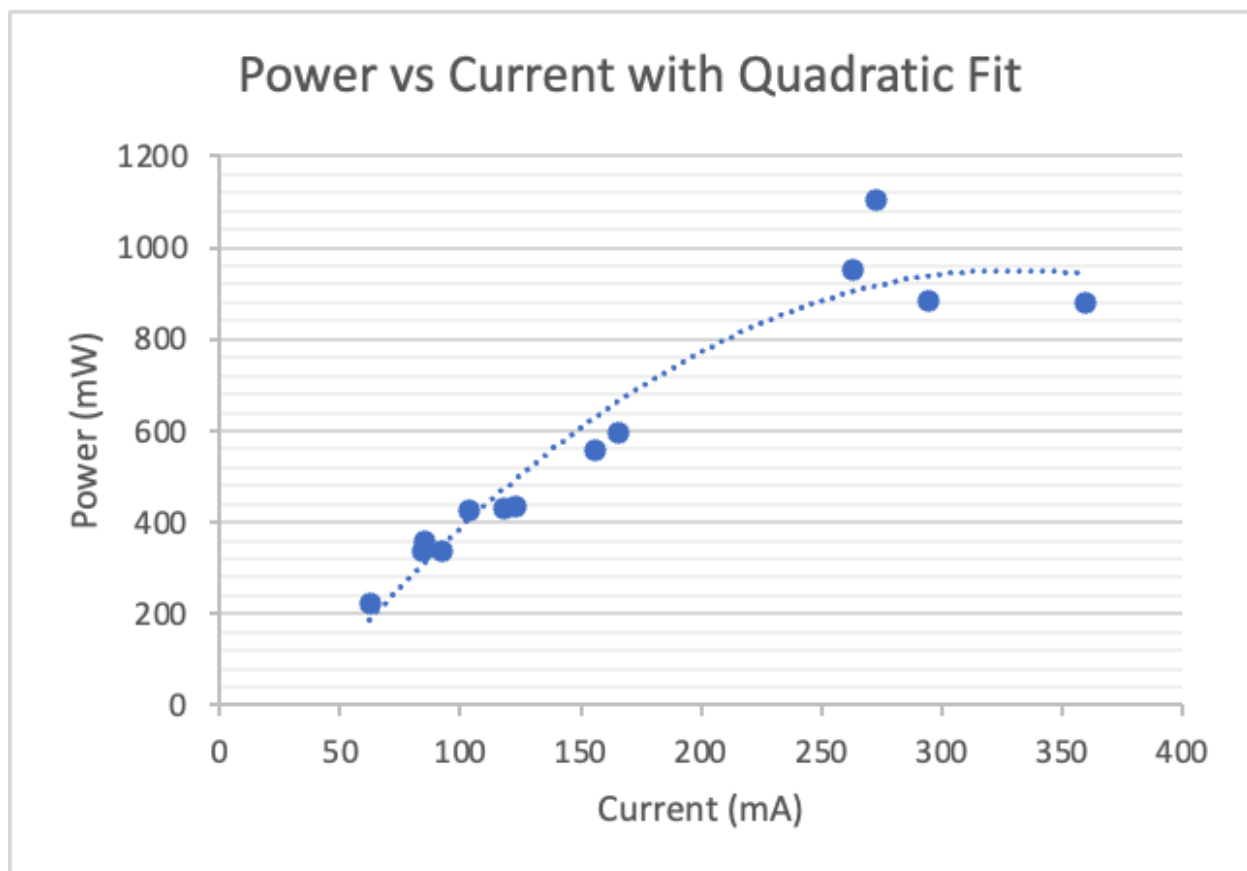


Linear Fit Results (Tower Stiffness)

- Equation:
$$\text{Load (N)} = 17.755 \times \text{Displacement (mm)} + 0.185$$
- Slope (Stiffness):
$$17.755 \text{ N/mm}$$
- R^2 value: 0.987 — indicating an excellent positive linear correlation between load and displacement.

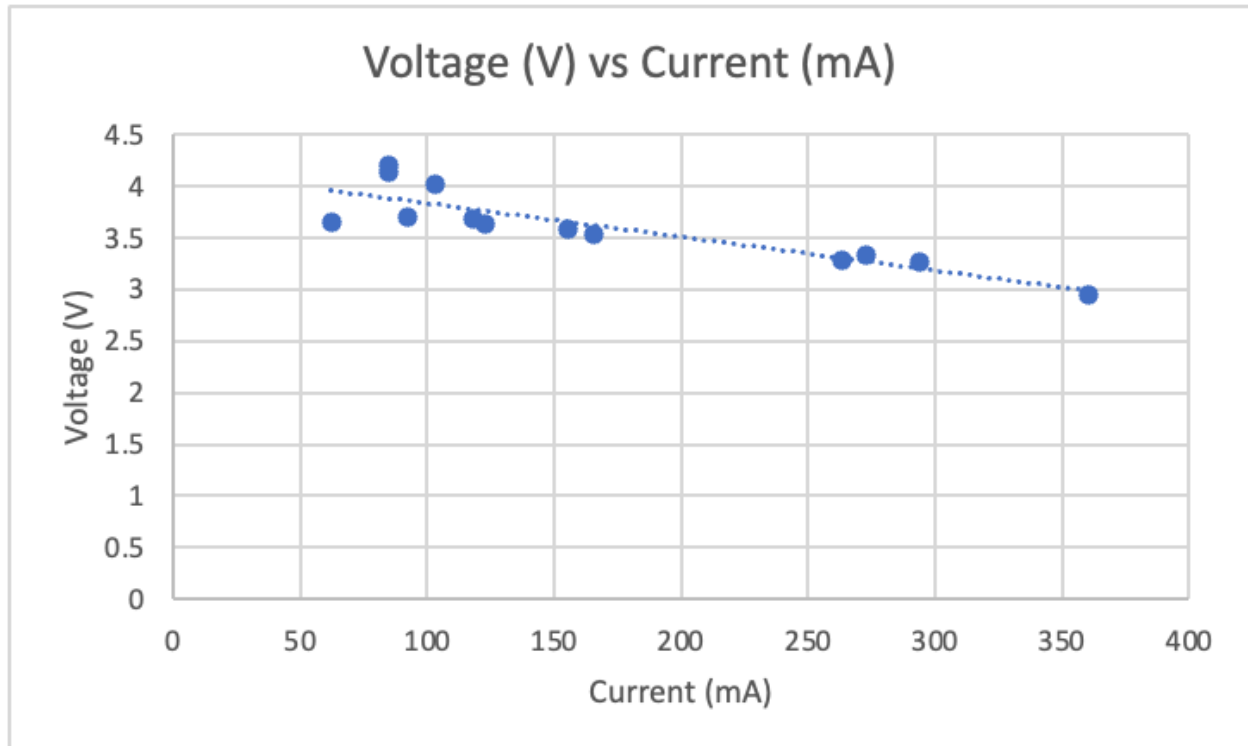
Plot 2: Power vs. Current

This plot shows the wind turbine's electrical power output (in milliwatts) as a function of current (in milliamps) under different electrical load conditions. The peak power output of 1105.0 mW occurred at 273.1 mA. This peak value was used in the efficiency calculation comparing experimental and theoretical power.



Plot 3: Voltage vs. Current

This plot shows the relationship between voltage (V) and current (mA) during the power generation test. It typically indicates a decrease in voltage as current increases due to load and internal resistance effects.



Theoretical Power (by Betz's limit)

$$P = \frac{1}{2} \times \rho \times A \times v^3 \times C_p$$

Where:

- ρ = air density $\approx 1.225 \text{ kg/m}^3$
- A = swept area of turbine blades (m^2)
- v = wind speed (m/s)

- $C_p = \text{Betz limit (maximum efficiency)} \approx 0.59$

Given:

- Wind speed: $23.7 \text{ mph} \approx 10.6 \text{ m/s}$
- Rotor diameter: $6 \text{ in} \approx 0.1524 \text{ m} \rightarrow \text{radius} = 0.0762 \text{ m}$
- Swept area: $A = \pi \times r^2 \approx 0.0182 \text{ m}^2$
- Air density: $\rho = 1.225 \text{ kg/m}^3$
- $C_p (\text{Betz limit}) = 0.59$

$$P_{\text{theoretical}} = \frac{1}{2} \times \rho \times A \times v^3 \times C_p = \frac{1}{2} \times 1.225 \times 0.0182 \times (10.6)^3 \times 0.59$$

$$\approx 7.8333 \text{ W} = 7833.4 \text{ mW}$$

Experimental Power:

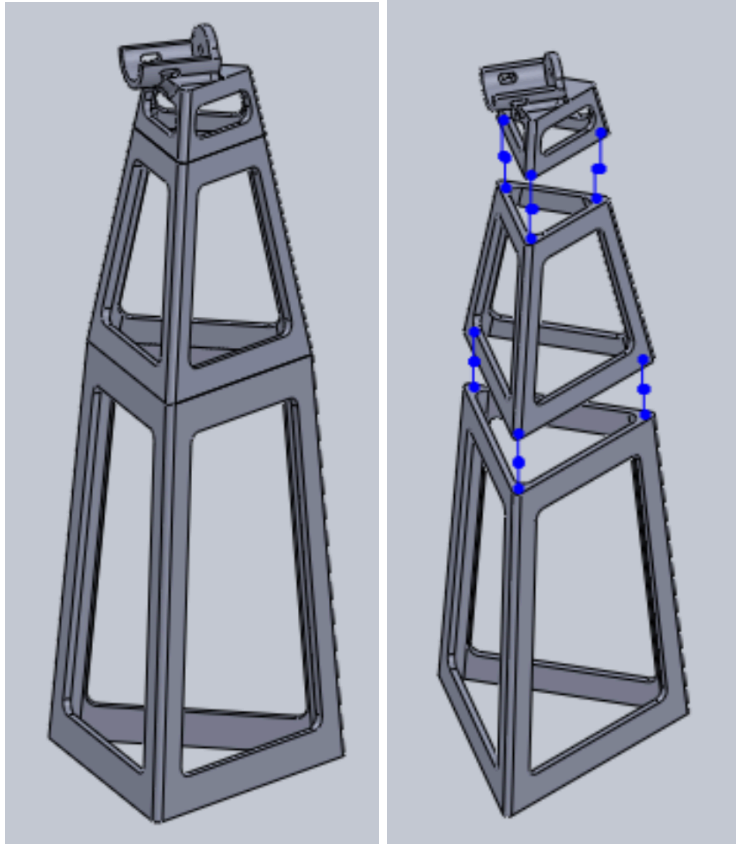
$$P_{\text{experimental}} = 1105.0 \text{ mW}$$

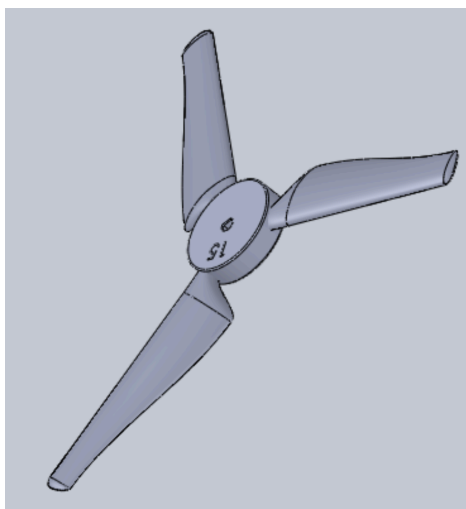
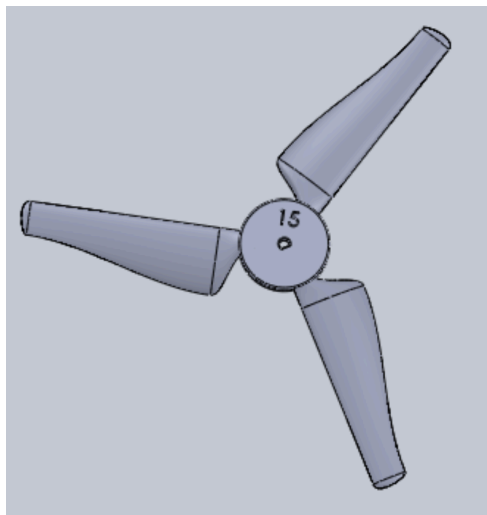
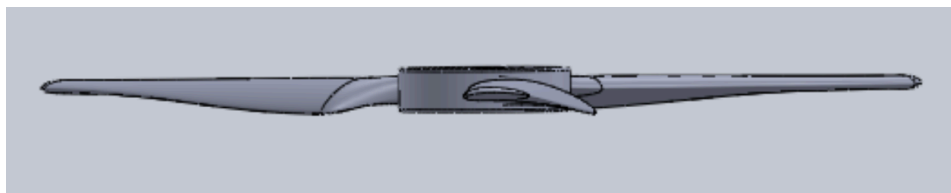
Efficiency:

$$\text{Efficiency} = \frac{P(\text{experimental})}{P(\text{theoretical})} = \frac{1105.0 \text{ mW}}{7833.4 \text{ mW}} \times 100\% = 14.1\%$$

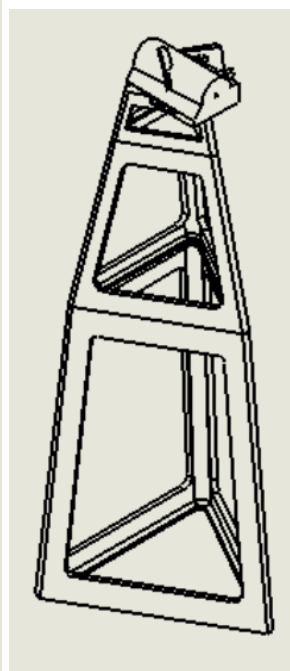
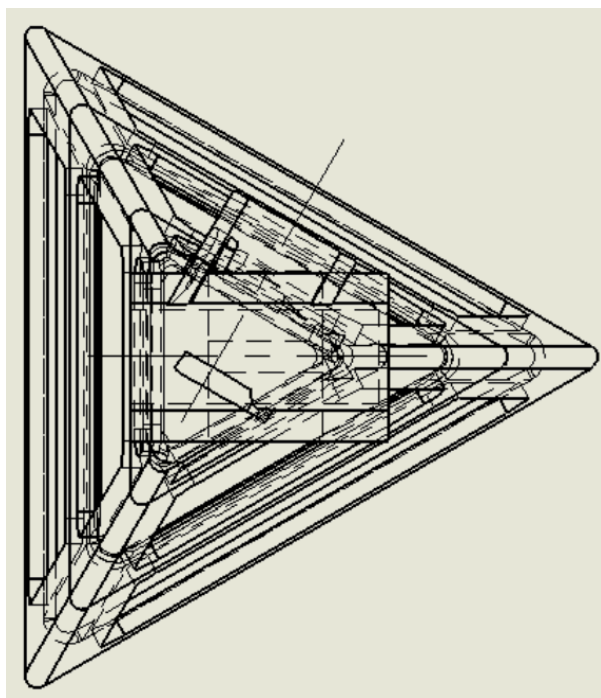
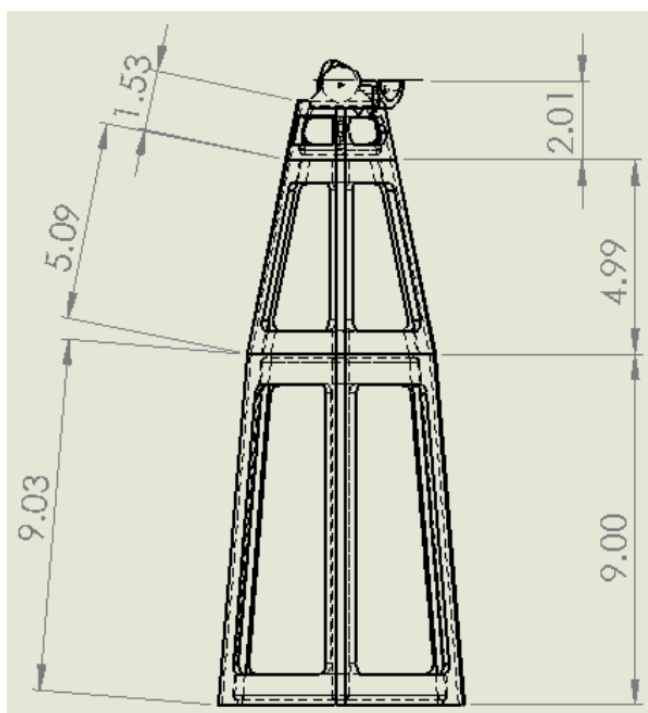
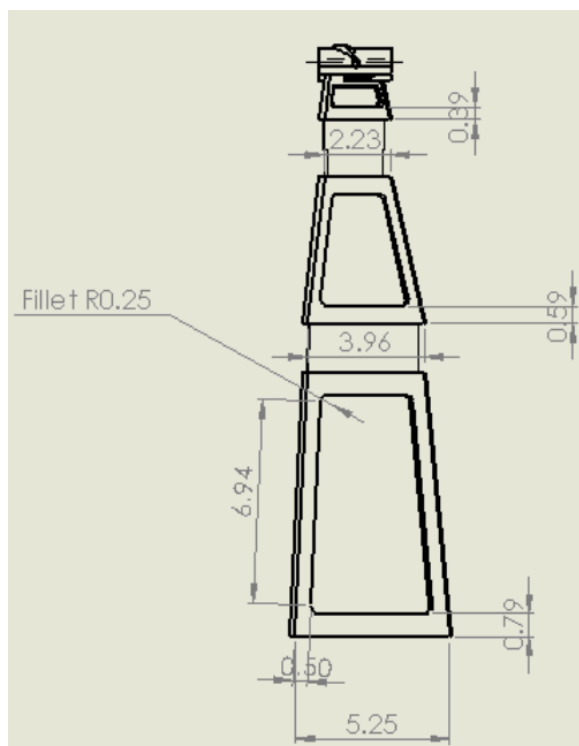
CAD Drawings

Tower Assembly and Exploded View

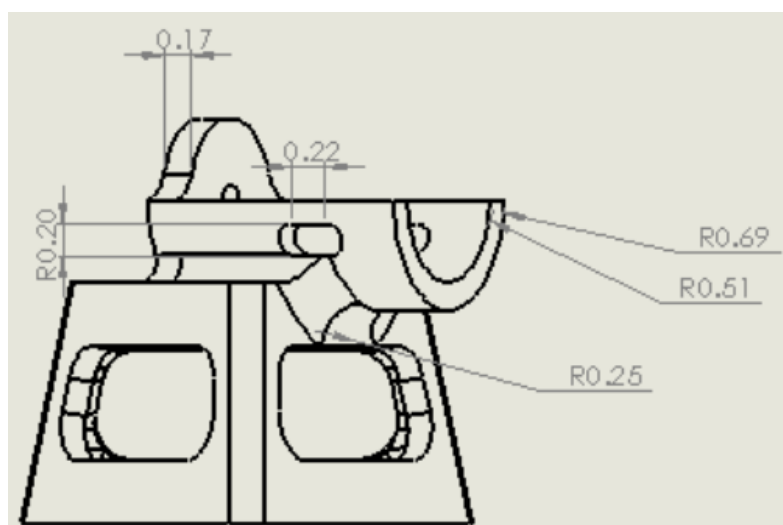
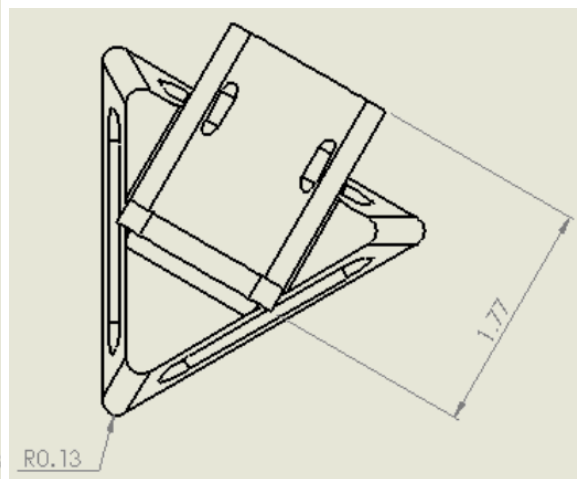
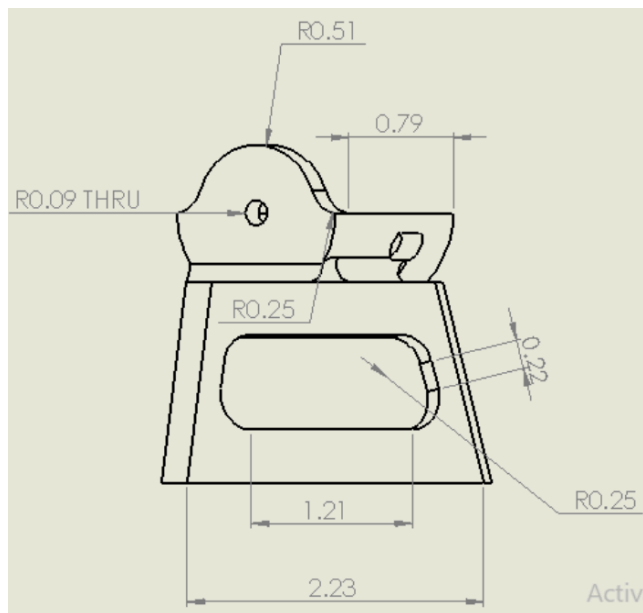


Blade/Turbine

Tower Drawings



Top Section/Motor Housing Drawing



Conclusions

This project aimed to design, fabricate, and evaluate a small-scale, 3D-printed wind turbine optimized for both power generation and structural performance. Operating under strict dimensional and volume constraints, our team applied core mechanical engineering principles to develop a functional, lightweight system capable of converting wind energy into electrical power while maintaining high stiffness-to-mass efficiency.

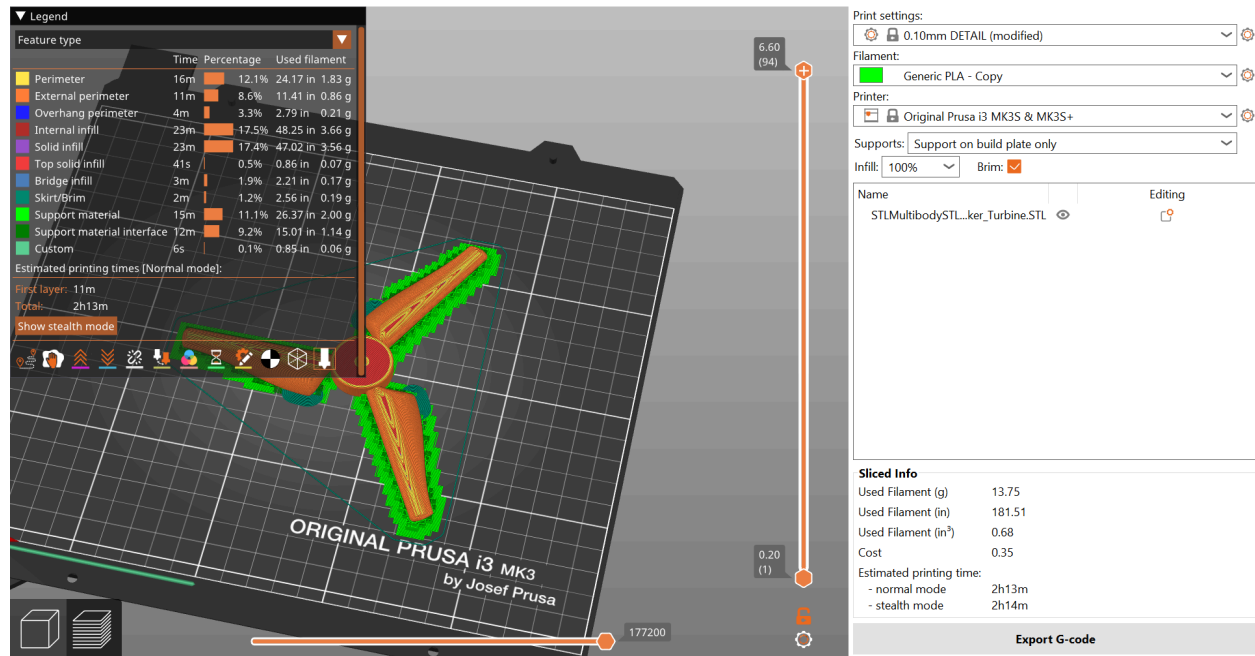
SolidWorks served as the primary design tool throughout the development process. It was used extensively for modeling all components, including the rotor blades, tower structure, and generator housing, allowing precise control over geometry, dimensions, and assembly interfaces. The integrated simulation environment in SolidWorks was also critical for structural validation. Finite Element Analysis (FEA) was performed on the tower to predict its deflection under vertical loading. The FEA model applied a distributed load to the top motor platform, simulating the weights used during physical testing. The resulting displacement, stress, and safety factor plots helped identify areas of high strain and informed modifications to reduce mass without compromising rigidity.

We successfully created a wind turbine with a 3-blade rotor inspired by the SG6043 airfoil, optimized for low Reynolds number conditions. The tower was structurally redesigned to meet the 17 in³ volume constraint while maintaining stability through truss-like features and modular assembly. Our experimental testing demonstrated a peak electrical power output of 1105 mW at 273.1 mA current, and the tower exhibited a stiffness of 17.755 N/mm with a strong linear correlation ($R^2 = 0.987$) between load and displacement. The tower assembly weighed at 305.3g. Although the system achieved only 14.1% efficiency relative to the theoretical Betz

limit, this was a valuable learning experience that highlighted real-world losses in conversion efficiency due to design trade-offs, manufacturing imperfections, and mechanical resistance.

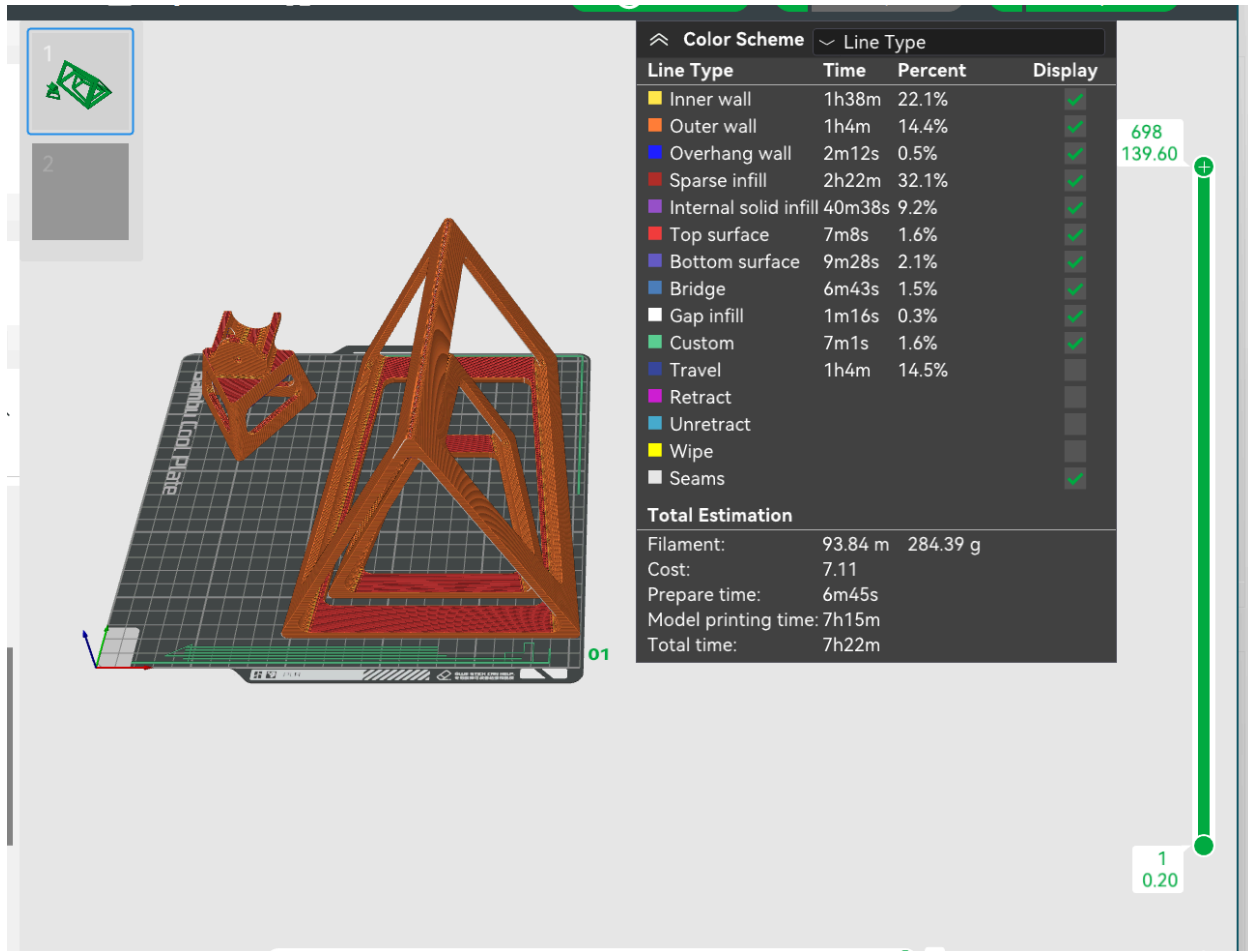
Beyond technical performance, this project provided practical insight into aerodynamics, structural mechanics, CAD modeling, additive manufacturing, and experimental validation. We gained experience in balancing design constraints with performance goals, refining manufacturing techniques, and troubleshooting prototype challenges. Overall, the project successfully fulfilled its objectives and provided a realistic context for applying core engineering concepts in renewable energy design.

Recommendations for Future Work



If given additional time and resources, several improvements could be made to enhance the performance, manufacturability, and overall efficiency of the wind turbine design. One key area for improvement is the tower geometry. Although the structure met our stiffness and displacement target of >8 N/mm, the final post-assembly mass slightly exceeded the 300g threshold, based on a measured value of $1200.3\text{g} - 895\text{g} = 305.3\text{g}$. If using the rubric's value of 910g for the baseplate, the final assembly would evaluate to $1200.3\text{g} - 910\text{g} = 290.3\text{g}$. If the base plate value was taken incorrectly, it is possible that we were under the 300g threshold.

We planned to print the final tower assembly using 295.2g of PLA filament in order to maximize stiffness while still complying with the mass limit. However, we could have reduced the weight to account for the gluing process and other unexpected sources of mass.



With more time, further FEA-informed revisions, like a topology plot, could have been performed to reduce unnecessary wall thickness while maintaining structural integrity, potentially incorporating internal lattice structures or tapered sections to optimize the displacement-to-mass ratio. We also intended to explore other designs that we initially proposed during the ideating lab; however, we ran out of time and had to prioritize meeting the volume and mass budget for the tower. Additionally, integrating shallow grooves or interlocking features along the bonding surfaces of the tower would have increased the surface area for glue contact and improved the strength of adhesive joints. Increasing the base surface area of the tower in contact with the baseplate would also improve overall adhesion and ease of assembly during the gluing process.



Manufacturing quality is another area with room for improvement. While we printed the blades and tower ourselves using a Prusa MK3S+ and Bambu X1C, respectively, and post-processed with a heat gun at 230 °C and wet sanding up to 600 grit, there were noticeable surface imperfections, including visible layer lines, warping, minor print defects, and remnants of support material, particularly on the blade.

Furthermore, using the original airfoil design resulted in uneven printing and required even more post-printing treatment. To avoid this, we thickened the blade as a whole, sacrificing parts of the airfoil's design for stability and thickness for smoother printing. These imperfections likely reduced aerodynamic efficiency, causing our turbine to reach 1105 mW instead of the goal of >2000 mW that we had set. In future iterations, we recommend printing with slower print speed, longer print jobs, and finer layer heights (0.05 mm instead of 0.10 mm) and applying post-processing techniques such as heat gun smoothing, epoxy coating, or high-grit sanding (beyond 600 grit) to achieve a smoother surface finish. These improvements could significantly reduce drag and enhance power output. Moreover, better printer calibration or higher-resolution printers could improve print fidelity, especially for the aerodynamics of the blade.

Material selection should also be revisited. PLA, while easy to print and dimensionally stable, is brittle and prone to bending and failure. Future testing with ABS or PETG could offer better durability and ductility, particularly under cyclic or impact loading.

Lastly, additional empirical testing, including physical testing of different printed blade designs, would help quantify how blade imperfections and tower deflection impact real-world performance.

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Appendices

Appendix A: Original Data Sheets

University of California
Mechanical Engineering Department – E26

Wind Turbine Tower stiffness test

Team member names:

Lab section 15 Total weight (grams) 1200.7g - 875g
(305.3g)

Deflection Measurements

(100g increments)

Data Points	Load (N)	Displacement (mm)	Observation
1	100	.08	
2	200	.11	
3	300	.16	
4	400	.20	
5	500	.26	
6	600	.32	
7	700	.35	
8	800	.41	
9	900	.47	
10	1000	.52	
11	1100	.60	
12	1200	.7	

University of California
Mechanical Engineering Department – E26

Wind Turbine Power Generation test

Team member names: 615 - Kina, Arjun, Krishna, Deyan, Sam, Arslan

Lab section 15 Wind speed (mph) 23.7
Measured at the rotor location

Power Measurements

Resistance (Ohms) (lead)

Data Points	Voltage (V)	Current (mAmp)	Power (mWatt)	Blade Rotational Speed (rpm)
1	3.65	62.20	225.2	6144
2	4.13	95.00	358.9	5974
3	4.21	84.60	336.1	6125
4	4.02	103.4	422.5	5334
5	3.70	92.4	337.4	5335
6	3.64	122.8	436.1	5249
7	3.68	118.4	431.5	5367
8	3.54	165.9	596.6	5280
9	3.58	155.6	558.9	5331
10	3.28	263.1	951.4	5073
11	3.34	243.1	1105	5136
12	3.26	294.1	883	5045
13	2.75	360.1	878.3	4714